

Yousuf Alami

Reading, UK | +44 7826 166562 | yousufalami99@gmail.com | LinkedIn: Yousuf Alami

PROFESSIONAL SUMMARY

MAIT (Manufacturing, Assembly, Integration & Test) Technician with hands-on experience assembling, integrating and testing high-reliability flight hardware across the full product lifecycle, from early-stage development through to flight-ready systems.

Experienced in delivering flight antenna systems now operating in orbit, with expertise in precision mechanical assembly, environmental testing, manufacturing documentation (MAPS), and ISO9001-controlled manufacturing environments requiring rigorous quality and traceability.

Proven experience delivering hardware from Breadboard (BB) and Engineering Model (EM) through Qualification, EQM, Proto-Flight Model (PFM) and Flight Model (FM), consistently maintaining exceptional build quality, repeatability and compliance with engineering requirements.

AREAS OF EXPERTISE

Technical Skills

MAIT (Assembly, Integration & Test), Flight Hardware Build, Precision Mechanical Assembly, Alignment & Fit Checks, Environmental Testing (Vibration, Thermal Cycling, Modal), RF Test Support, Laser Scanning Inspection, MGSE & EGSE Operation, MAPS Documentation (Authoring & Execution), Engineering Drawings, Fault Finding, Traceability & QA (ISO9001)

Operational Strengths

Full Lifecycle Hardware Delivery, Batch & One-Off Builds, Build Ownership & Responsibility, Fast-Paced Development Environments, System Integration, Customer-Facing assembly, integration and test Support, External Site Work, Problem Solving Under Schedule Pressure, Process Standardisation

PROFESSIONAL EXPERIENCE

MAIT Aerospace Technician (Full-Time)

Oxford Space Systems — Sep 2023 – Present

Didcot, England, United Kingdom (On-site)

- Assemble, integrate and test high-reliability flight hardware within an ISO 7 cleanroom environment.
- Deliver multiple flight antenna systems now operating in orbit, including both batch production and one-off flight hardware.
- Integrate customer-furnished equipment, including star trackers and magnetometers, into system-level assemblies.
- Build, integrate and functionally verify multiple types of Hold-Down & Release Mechanisms (HDRMs) supporting deployment-critical space systems.
- Perform precision alignment, fit checks and verification activities to ensure compliance with engineering requirements.
- Diagnose and resolve manufacturing and integration issues to maintain programme schedules.
- Author and maintain Manufacturing & Assembly Procedure Specifications (MAPS) for sub-assemblies and top-level flight hardware.
- Review engineering drawings, BOMs and manufacturing documentation to ensure build accuracy and configuration control.
- Execute environmental and functional testing, including thermal cycling, modal testing, deployment testing, RF test support, vibration test support and laser scanning.
- Assemble and operate Mechanical and Electrical Ground Support Equipment (MGSE/EGSE) for safe integration and testing of flight hardware.

- Maintain full product traceability and quality assurance in accordance with ISO9001 processes and formal manufacturing documentation.
 - Deliver hardware across the complete development lifecycle, from Breadboard (BB) and Engineering Model (EM) through Qualification, EQM, Proto-Flight Model (PFM) and Flight Model (FM), while consistently meeting demanding quality, schedule and compliance requirements.
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MAIT Aerospace Technician (Apprenticeship)
Oxford Space Systems — Sep 2019 – Sep 2023
Didcot, England, United Kingdom (On-site)

- Completed a Level 3 Manufacturing Engineering Apprenticeship while supporting the assembly, integration and testing of spaceflight hardware.
 - Developed core skills in precision mechanical assembly, engineering drawings, manufacturing processes and cleanroom working practices.
 - Supported environmental testing, functional verification and hardware integration activities across multiple development programmes.
 - Gained practical experience in manufacturing documentation, product traceability and ISO9001 quality assurance processes within a regulated aerospace environment.
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KEY PROJECTS & FLIGHT HERITAGE

Flight Heritage: Contributed to the delivery of **23+ flight hardware units currently operating in orbit**, with additional flight models delivered for future missions and supporting multiple programmes from early development through to flight.

S-Band Isoflux Helical Antenna Programme (Oxford Space Systems → Astro Digital)

- The build+test technician responsible for antenna hardware throughout the full development lifecycle, from Breadboard (BB) and Engineering Model (EM) to Flight Models (FM).
- Built **four Flight Models**, with **two successfully launched and operating in orbit, one delivered for a future mission, and one retained as a flight spare**.
- Performed precision assembly, integration, alignment and verification of high-power RF deployable RF systems while supporting the transition to controlled flight hardware using MAPS.

Lacuna Space Helical Antenna Programmes (Oxford Space Systems × Lacuna Space)

- The build+test technician responsible for batch production and one-off flight antenna builds across multiple programmes.
- Built **20+ flight antenna units** successfully launched and operating in orbit, including two batch production campaigns that contributed to Lacuna Space's **10/10 in-orbit deployment success**.
- Developed and refined MAPS documentation to improve manufacturing repeatability, process standardisation and full product traceability.

Helical Magnetometer Boom Programme (Oxford Space Systems × IOTA Technology × AAC SpaceQuest)

- The build+test technician responsible throughout the full development lifecycle, from Breadboard (BB) and Engineering Model (EM) to Engineering Qualification Model (EQM) and Proto-Flight Model (PFM).
- Built and integrated the Proto-Flight Model, supporting customer hardware integration, system verification and launch preparation.

- Provided on-site customer support during testing at RAL Space and the NASA Goddard Magnetometer Test Facility, with the flight hardware successfully launched and currently operating in orbit.

Offset Reflector Antenna – First Flight Model (FM) (*Oxford Space Systems*)

- Key member of the cross-functional team delivering the company's first Flight Model offset reflector antenna.
- Supporting assembly, integration and verification while resolving manufacturing challenges during the transition from development to flight hardware.
- Programme in final build and verification ahead of planned Q4 2026 delivery.

VHF Isoflux Helical Antenna (ECHOES Project) (*Oxford Space Systems* → *Startical*)

- Build technician from initial concept through to Engineering Model (EM) development.
- Supported prototype development, design iteration and manufacturing process optimisation to enable progression towards flight hardware.
- Contributed to the development of hardware that was subsequently matured into a flight-qualified system and successfully deployed in orbit.

Wrapped Rib Antenna – 5 m Engineering Qualification Model (EQM) (*Oxford Space Systems*)

- Build technician supporting the assembly and integration of a 5 m Engineering Qualification Model.
- Integrating a large deployable space structure within a high-precision, fully traceable manufacturing environment.
- Supporting qualification activities ahead of progression to future flight hardware.

EDUCATION

HNC Mechanical Engineering (Distinction)

Oxfordshire Advanced Skills — 2021–2023

Level 3 Extended Diploma in Manufacturing Engineering (Distinction)

Oxfordshire Advanced Skills — 2019–2023

A Levels – Design Technology, English Language, Sociology, Economics

Maiden Erlegh Sixth Form — 2016–2018

PROJECT REFERENCES

- Astro Digital – S-Band Isoflux Helical Antenna Deployment
- Lacuna Space – 10/10 In-Orbit Deployment Success
- MagQuest / IOTA Technology – Io-1 Mission Deployment
- Startical – ECHOES VHF Helical Antenna
- Oxford Space Systems – Flight Heritage

Project Portfolio & Public Flight Heritage available on request or via the attached portfolio.